

Dynamic AI Exposure (DAX): pre-registered design memo v1

Status: PI DEFAULTS APPROVED 2026-08-06 — EVIDENCE PENDING; not pre-registered

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Gate: DAX-GATE1-memo

Outcome-data status: SEALED until the signed tag v1.0-preregistered

This memo translates the DAX proposal and Amendment v1.1 into an executable design. The PI approved all 17 bracketed [PI-DECISION] defaults and all six non-numeric confirmations on 2026-08-06. They are frozen design choices, but the tag remains prohibited until the evidence checklist, power task, independent red-team, and rendered review are complete.

0. Binding scope and feasibility conditions

The estimand concerns the U.S. occupation-level *full-task displacement frontier*: the wage-bill share of tasks for which AI substitution becomes privately cost-effective. It is not a net general-equilibrium employment effect and does not treat technical feasibility as realized adoption.

The signed 2026-07-10 feasibility decision binds this memo:

1. W4 capture of accessible historical model snapshots must finish before the applicable 2026-10-23 and 2026-12-11 shutdown dates.
 2. Retired vintages enter only through cited stand-ins, stand-in uncertainty enters the EIV analysis, and gpt-4.5-preview is excluded because no qualified stand-in was filed.
 3. GDPval is referenced by task ID for internal research. No GDPval task text or derived task content enters W10a unless redistribution rights are clarified.
- Primary task bundles, wages, and weights use 2021 ONET/OEWS vintages. A 2019 wage baseline and annually updated ONET bundles are robustness/decomposition variants, not replacements for the frozen primary index.

1. Event registry

1.1 Inclusion rule

An event enters the registry only if it changes at least one of: accessible model capability, input/output/reasoning-token price, or the set of model snapshots available for measurement. Every date and price requires two locators. Announcement-only events without API availability are recorded but do not receive an exposure dose until the API-effective date.

1.2 Machine-readable registry and source-complete core events

The frozen registry source is dax/memo/event_registry_v1.csv. It records analysis status separately from evidence status: a candidate is chronology, not permission to estimate an event. python dax/memo/validate_event_registry.py fails if IDs or dates are malformed, either locator is absent, sources repeat, or an event marked eligible lacks verified status. The validator does not pretend to read source content; rows marked pending_second_date_locator remain ineligible until a second source independently dates API availability.

Event ID	API-effective date	Classification	Evidence 1	Evidence 2	Status
GPT4_LAUNCH	2023-03-14	capability	OpenAI, GPT-4	OpenAI, GPT-4 technical report	date verified; price-history row pending W2
GPT4O_LAUNCH	2024-05-13	capability + price	OpenAI, Hello GPT-4o	OpenAI, GPT-4o product release	lead price event; official release states API cost was 50% lower than GPT-4 Turbo
O1_PREVIEW	2024-09-12	capability	OpenAI, Introducing o1-preview	OpenAI API changelog (September 2024)	date verified
GPT41_LAUNCH	2025-04-14	capability + price	OpenAI, Introducing GPT-4.1 in the API	OpenAI API changelog (April 2025)	date and launch prices verified

The current registry contains 21 dated chronology rows through 2026-08-06: four currently eligible core events, sixteen candidates, and the binding gpt-4.5-preview exclusion. Four candidates are explicitly pending a second independent date locator. Candidate status is resolved before Gate 1; W5 later applies the signed dose threshold without adding or deleting chronology rows. The mechanically fillable event-by-event table is frozen in event_table_shell_v1.csv. W5 may fill only its blank window, crossing-count, dose-distribution, and effective-weight fields after applying Decisions 1–3; it may not change event IDs, dates, thresholds, or column definitions. At the Gate-1 freeze, any candidate still lacking two independent dated locators or a source-complete dated price row is mechanically reclassified source_failed and excluded from the primary event set. It remains in the chronology with the failed source fields and exclusion reason; it cannot be restored after outcomes open except through a dated deviation memo. This rule includes every unresolved conflict_b: W2 must select the price row supported by two independent dated locators and archive the conflicting rows, or the event becomes source_failed; averaging or choosing the lower price is prohibited. If W4 cannot recover either a qualifying historical snapshot or a feasibility-approved cited stand-in, the event is likewise excluded as measurement_failed and retained only in the descriptive chronology. Registry price status relative_price_verified means that two locators verify a relative price change but W2 has not yet attached the absolute pre/post price rows needed for dose construction; it is therefore not source-complete at Gate 1 unless W2 supplies those rows. pending_w2 has no verified price change, and n_a is allowed only for a capability-only event whose price is unchanged.

1.3 Registry completion queue

Candidate rows now cover GPT-4 Turbo/DevDay, GPT-4o mini, full o1, o3-mini, o3/o4-mini, GPT-5-family launches through GPT-5.6, and located price changes. They remain non-filed analysis events until the CSV records two date locators and W2 attaches dated prices. W2 emits the source-complete price-history panel; W1 imports its event IDs without changing the rules below.

The current official deprecations page contains both a historical 2025-09-26 row saying `gpt-4-1106-preview` shut down on 2026-03-26 and a newer 2026-04-22 row listing an October 23, 2026 shutdown for the same slug. The signed feasibility adjudication uses the newer row for capture planning. The memo preserves both locators as a provenance conflict; W4 must test actual availability before budgeting the row.

[PI-DECISION 1] Event eligibility default. Require at least one percentage point of occupation wage-bill dose at the median occupation for an event to be included in the primary stacked analysis. Smaller events remain in the DAX panel and descriptive chronology but are not separately estimated. This avoids nearly unidentified event coefficients while leaving the index complete.

2. DAX construction objects fixed for the design

For task t , occupation o , month m , mapping k , cost variant v , deployment parameter δ , and failure-cost rule g , define

$$pi_eff = 1 - \delta * (1 - pi)$$

and crossing indicator

$$A_tom = 1[c/pi_eff + f*(1-pi_eff)/pi_eff < w].$$

DAX_{om} is the frozen-2021 wage-bill share of occupation tasks with $A_{tom}=1$. ΔDAX_{oe} is the share newly changing from zero to one at event e . A task cannot cross twice in the primary monotone chronology. If a measured capability regression or price increase reverses feasibility, the reversal is retained as a signed change in a separate non-monotone diagnostic and does not erase its first crossing.

Primary grid:

- mapping: GDPval, Tolan-style, Eloundou-style;
- cost: API list, $0.6 * list$, open-weight marginal;
- δ : 1.0, 0.8, 0.6;
- failure multiple by O*NET consequence tier: 0.25, 1, 4 times task wage;
- failure sensitivity: zero, half, primary, and double tier multiples;
- cost accounting: billed tokens and completion-only effective tokens for

reasoning models;

- wage vintage: 2021 primary, 2019 alternative.

The flip rate is the share of occupation-event cells whose crossing status changes anywhere across the failure-cost grid. The already-proposed trigger is strictly greater than 20%. Crossing this trigger halts W5 before outcome work and returns the threshold design to the PI; it does not authorize choosing the best-performing grid point.

3. Stacked common-event dose-response protocol

3.1 Unit, dose, and comparison

The analysis unit is person-month in IPUMS-CPS, with DAX doses assigned through the employment-weighted many-to-many CPS occupation to O*NET-SOC crosswalk. The event-specific treatment is continuous ΔDAX_{oe} . The comparison set for event e contains occupations in the same frozen static-exposure decile with zero crossing dose at e and no crossing inside that event's clean window.

3.2 Four stacking parameters

[PI-DECISION 2] Window default. Convert each API-effective date to its CPS calendar month and start with event time $[-6, +6]$. If multiple eligible rows fall in the same calendar month, replace them in the adjacency set with one compound labor-market event whose event-time origin is the first day of that CPS calendar month. Compute its dose once from the joint pre-to-post state, so a task changed by multiple components contributes at most one crossing; retain the component rows and exact API dates only in descriptive tables. For each remaining monthly event, retain a month on the pre side only when its integer month distance to the focal event is strictly smaller than its distance to the previous eligible event, and retain a month on the post side only when its distance to the focal event is strictly smaller than its distance to the next eligible event. A calendar month exactly equidistant between two events is excluded from both windows. Intersect these sets with $[-6, +6]$, then retain an event only if at least three uncontaminated pre months and three uncontaminated post months remain. For example, two April component rows and two December component rows become April 1 and December 1 compound origins; the four components are removed from adjacency. The origins are eight month indices apart, so August is the tied midpoint and is excluded; April keeps May–July and December keeps September–November on their adjacent sides.

The compound-state algorithm is mechanical: order components by exact API-effective timestamp and then `event_id`; define the pre state immediately before the first component; sequentially apply each verified capability, snapshot-availability, and price change; and define the post state immediately after the last component. Recompute A_{tom} on those two states for every task and set the compound ΔDAX to the wage-bill share changing from zero to one. A one-to-zero change remains only in the signed non-monotone diagnostic.

[PI-DECISION 3] Minimum occupation dose default. Define a treated occupation-event cell as $\Delta DAX \geq 0.01$; retain smaller positive doses in continuous-dose estimation but exclude them from treated-bin summaries. One percentage point is economically interpretable and limits classification noise near zero.

The Decision-1/Decision-3 order is fixed. W5 first computes raw, unthresholded Δ_{DAX} for every source-verified candidate using the frozen task panel, before windows or treated bins are constructed. Decision 1 is then evaluated on the unweighted median of those raw occupation-level wage-bill shares across mapped CPS occupation codes. Only events passing that eligibility rule enter the adjacency set used by Decision 2. Decision 3 is applied last and only to treated-bin summaries; values in $(0, 0.01)$ remain in the continuous regressor. Because Decisions 1 and 3 use the same weak threshold, every evaluable event passing Decision 1 necessarily has at least half of its mapped CPS occupations in the treated bin under the registered sample-median convention. An event with no mapped occupation has no defined median, fails Decision 1, and remains descriptive; no threshold is lowered. All Decision-1 passers enter the continuous-dose regression.

[PI-DECISION 4] Dose accumulation default. The primary event coefficient uses the event increment Δ_{DAX_oe} , not the cumulative DAX level. Current DAX level and prior cumulative dose enter as pre-event state controls. This isolates the new information at the event while respecting that exposure stock persists.

[PI-DECISION 5] Earlier-crossers default. Earlier-crossing occupations stay eligible if they receive a new positive increment at the current event and have no other crossing inside the current clean window. They cannot serve as zero-dose controls. Occupations with only prior exposure may serve as controls within static decile after conditioning on pre-event DAX level. Report a robustness sample excluding all prior crossers.

Each event contributes person weights normalized to sum to one within event, so large CPS months or longer surviving windows do not mechanically dominate the stack. Standard errors cluster by the original CPS occupation code.

4. Identifying conditions and estimands

IC-1 — Conditional strong parallel trends for a continuous dose. For every eligible event, event time, and two doses in the common support, the expected change in untreated potential outcomes is identical across doses conditional on frozen static-exposure decile, occupation fixed effects, calendar-month fixed effects, industry-by-month fixed effects, occupation interest-rate sensitivity, pre-event DAX level, and the registered baseline covariates.

IC-2 — No event-window dose-correlated interference. Conditional on the same controls, spillovers from high-dose occupations to comparison occupations do not vary with event dose inside the clean window. Mobility-network spillover estimates are auxiliary and do not repair violation of IC-2.

IC-3 — Mapping timing exclusion for secondary IV. Conditional on controls, errors in Tolan and Eloundou doses are independent of the GDPval mapping error and untreated outcome innovations. Shared structural errors—such as all mappings equating benchmark performance with production success—violate this condition and are not removed by IV.

Primary estimand: the change in employment probability and unconditional weekly hours associated with a 0.10 increase in event-specific DAX dose, averaged over eligible events and the observed common-support population. Event-specific and binned nonparametric estimates accompany the pooled linear dose response.

5. Mapping hierarchy and measurement quality

GDPval is primary. Tolan and Eloundou mappings are independent robustness constructions. All task matches retain similarity, coverage, confidence, adjudication status, and a top-quartile match-quality flag. Unmatched tasks remain explicit and their wage-bill share is reported.

The pre-existing survive-all-three rule is binding: the point-estimate sign is consistent across all three mappings and the relevant test rejects at the 10% level under at least two mappings.

[PI-DECISION 6] Mapping tiebreaker default. If all signs agree but sampling precision differs, report GDPval as primary and the median standardized point estimate across mappings as the descriptive synthesis. The tiebreaker never overrides a sign conflict or the survive-all-three failure rule.

[PI-DECISION 7] Human-validation targets. Audit 10% of mapping-C annotations stratified by occupation family, score decile, and ambiguity flag. Require weighted Cohen's kappa of at least 0.70 and at least 90% agreement on the binary crossing-relevant label before W5. Failure returns the rubric for redesign before index construction.

6. Index validation and behavioral first stage

6.1 Public index validation

[PI-DECISION 8] Convergent-validity threshold. The primary GDPval/list/ $\Delta_{DAX}=1$ DAX level must have occupation-level Spearman rank correlation of at least 0.50 with the frozen Felten-Eloundou-Webb ensemble at one or more pre-registered benchmark months, with Kendall correlation reported. This is a moderate convergence requirement: lower suggests construct failure, while a near-one requirement would eliminate the intended dynamic/cost distinction.

Event alignment, capability-versus-price decomposition, crossing-order stability, and comparison with published usage indices are reported without post-hoc pass thresholds except those explicitly added by PI signature.

6.2 Usage-share first stage

The named human team alone runs frozen code on real OpenAI aggregates. The agent-developed pipeline sees synthetic data only.

The first-stage unit is frozen O*NET task-or-IWA cell by event by calendar month. For cell i and month t , within-month usage share is the number of classified eligible ChatGPT messages assigned to i divided by all classified eligible messages in the same negotiated product population and calendar month, before cell suppression. For event e , the pre-event mean is the simple arithmetic mean of that share over every surviving clean pre month for which the cell is observed; the complete-case rule in Section 8 requires observation in all required pre and post months. The normalized outcome is $100 \times (\text{share}_{iet} / \text{premean}_{ie} - 1)$. If $\text{premean}_{ie}=0$, the normalized outcome is undefined and the cell is excluded from the normalized first stage rather than regularized. Its raw share path is reported as an auxiliary diagnostic, along with the excluded task-mass share.

[PI-DECISION 9] Minimum behavioral effect. Success requires the pooled post-crossing coefficient on that normalized outcome to be at least 5, meaning a 5% relative usage-share increase, with its 90% confidence interval excluding zero. Event-time coefficients remain required diagnostics. Asking-to-Doing composition is a mechanism outcome; default materiality is a two percentage-point increase in Doing share, reported with uncertainty but not a separate gate.

A positive first stage supports dateable translation from feasibility to behavior. A null does not disprove technical feasibility; it localizes adoption away from measured ChatGPT message channels or indicates that unit economics is not the binding adoption margin.

7. CPS sample, outcomes, and power-analysis specification

Population: U.S. persons ages 22–25 in monthly IPUMS-CPS, November 2021 through the latest frozen month, using person weights. Dose assignment occurs at the CPS occupation code after employment-weighted crosswalking to O*NET-SOC. For each event stack, a person's reference occupation is the most recent valid CPS occupation observed for the same `CPSIDP` strictly before the event month and no more than 15 calendar months earlier; that reference is held fixed throughout the stack. This preserves unemployed respondents and prevents an event-induced occupation change from redefining treatment. A person with no valid pre-event occupation observation is excluded from that event stack, and the exclusion count, weighted share, and lookback- duration distribution are reported. A registered robustness check caps the lookback at nine calendar months.

Primary outcomes:

1. employment indicator;
2. unconditional usual weekly hours, coded zero for non-employed persons.

Secondary outcomes: log hourly wage among employed persons and conditional weekly hours. Auxiliary outcomes: occupation switching, unemployment reason, and search duration. The college-attainment split is secondary, receives its own power table, and cannot support headline claims.

[PI-DECISION 10] Multiplicity default. Control family-wise error across the two primary outcomes with Holm at two-sided 5%. Report unadjusted confidence intervals plus adjusted p-values. Apply Benjamini-Hochberg at 10% separately to secondary and auxiliary families.

The W1 power task must simulate the exact stacked structure using pre-event CPS moments only. It must preserve occupation cluster sizes, person weights, crosswalk dose dispersion, event windows, and the observed covariance between employment and hours. Simulated effects are injected after event time zero; the code cannot estimate treatment effects on real post-event outcomes before Gate 1.

[PI-DECISION 11] Power standards. Use two-sided alpha 0.05 and 80% power. Report MDEs for a 0.10 DAX dose and benchmark them against the 13% payroll decline cited in the proposal. The preferred design is adequately powered only if the employment MDE is no larger than 6.5 percentage points or one-half of the corresponding baseline employment gap, whichever is smaller. Failure does not authorize sample/specification search; it triggers the proposal's informative-power limitation and reconsideration by the PI. For unconditional hours, also report the weighted pre-event high-dose versus zero-dose baseline gap. Its 80% MDE for a 0.10 DAX increase must be no larger than two hours per week or one-half of that baseline gap, whichever is smaller. If either co-primary outcome fails its standard, that outcome is retained and reported as informative but cannot receive confirmatory interpretation.

Every output reports CPS-to-ONET dose dispersion. **[PI-DECISION 12]** Flag a CPS code as low-quality when the weighted within-code standard deviation of ONET dose exceeds 0.10 or the maximum mapping weight is below 0.50. Re-estimate on the complement as a registered robustness check.

8. Suppression, privacy, and minimum estimability

Suppressed OpenAI aggregate cells are missing by design and are never filled with zero, midpoint, model prediction, or neighboring-cell values. Primary first-stage estimation uses only cells observed in every required event-time period. An inverse-observation-probability weighted sensitivity analysis may use only publicly described suppression predictors and is labeled secondary.

[PI-DECISION 13] Minimum estimability default. Interpret the first stage only if: (a) observed cells cover at least 70% of the wage-bill-weighted crossed task mass; (b) at least 50 occupation/task cells remain; (c) at least three eligible events remain; and (d) no single event contributes more than 50% of the effective estimation weight. For event e , that weight share is $\frac{\sum_i w_{ie} * x_{\tilde{ie}}^2}{\sum_j \sum_i w_{ij} * x_{\tilde{ij}}^2}$, where w is the event-normalized CPS person weight and $x_{\tilde{}}$ is the primary continuous dose-by-post regressor residualized on the complete frozen nuisance design. Otherwise report selection and MDEs without an effect interpretation.

Only the PI and named RA may open or run code on real usage aggregates. The repository stores schemas, synthetic data, frozen code, and aggregate cleared diagnostics only—never cell-level NDA data or derivatives.

9. Econometric implementation and diagnostics

The primary pooled model interacts continuous event dose with event-time indicators and includes occupation, event-stack, calendar-month, and industry-by-month effects, plus registered controls. Every linear estimate is paired with nonparametric dose bins determined from the pre-event dose distribution and frozen before outcomes are opened.

Required diagnostics:

- joint test of all pre-event dose coefficients;
- binned dose-response plot for every primary parametric result;
- common-support and leverage plots by event;
- with/without industry-by-month fixed effects;
- within-static-decile dose/industry-composition correlation;
- leave-one-event-out estimates;
- Rambachan-Roth sensitivity at $M\text{-bar}$ 0.5, 1, and 2, plus the

relative-magnitudes restriction; $M\text{-bar}$ 1 is headline;

- FOMC dates, between-event pseudo dates, permuted mappings, and

backward-shuffled price histories as placebos.

[PI-DECISION 14] Pre-trend default. A design passes the conventional pre-trend diagnostic only if the joint pre-event test has $p \geq 0.10$ and no single pre-period adjusted confidence interval excludes zero after Holm correction. This diagnostic is not proof of IC-1; HonestDiD sensitivity remains mandatory regardless of the test result.

10. EIV, cross-mapping IV, and deployment calibration

10.1 Errors-in-variables simulation

For each mapping, perturb task success probability using distributions calibrated to human disagreement and item-level score dispersion, redraw at least at task and occupation-correlated levels, and rederive crossings.

[PI-DECISION 15] EIV acceptable bounds. Before outcome work, require median absolute crossing-date shift no greater than one month, 90th-percentile shift no greater than three months, no more than 10% of occupation-event cells changing dose bin, and implied linear attenuation no worse than 0.80. In each EIV draw, construct the exact frozen stack and residualize both the unperturbed dose-by-post regressor x_{true} and perturbed regressor x_{obs} on the same nuisance matrix with the same event-normalized weights. The draw-specific attenuation factor is the weighted slope $\text{Cov}_w(x_{\text{obs_tilde}}, x_{\text{true_tilde}}) / \text{Var}_w(x_{\text{obs_tilde}})$: the coefficient obtained from outcome-on-observed-dose when the true coefficient is one and no sampling noise is added. The 0.80 bound applies to the median factor across draws; report its 5th–95th percentile interval. Exceeding any bound does not permit selecting a quieter mapping; it triggers redesign or an explicitly weakened estimand.

10.2 Cross-mapping IV

Instrument GDPval_{DeltaDAX} separately and jointly with Tolan and Eloundou doses. Report reduced forms, first stages, overidentification diagnostics, and the IV/OLS ratio as a reliability diagnostic.

[PI-DECISION 16] Weak-instrument rule. Interpret IV magnitudes only when the heteroskedasticity/cluster-robust effective first-stage F statistic is at least 10. Otherwise report weak-instrument-robust intervals and treat IV as uninformative.

10.3 Post-first-stage delta calibration

A raw ratio of observed to predicted usage jumps is not, by itself, a valid estimate of delta, because delta changes π_{eff} , crossing sets, and predicted event jumps nonlinearly. DAX v1.1 therefore uses the pre-registered minimum-distance estimator

$$\text{delta_hat} = \underset{d \in [0,1]}{\text{argmin}} \sum_e \text{weight}_e * (J_{\text{obs}_e} - J_{\text{pred}_e}(d))^2,$$

where $J_{\text{pred}_e}(d)$ is recomputed through the full crossing pipeline and weights are inverse estimated variances of observed jumps. Bootstrap resamples the eligible usage cells and events. This calibration is post-first-stage and cannot change this paper's primary grid $\{1.0, 0.8, 0.6\}$. Estimate each jump variance with a pre-registered two-way pigeonhole bootstrap: 1,000 draws independently resample eligible task/IWA cells and eligible events with replacement, rerun the frozen first stage, and use the sample variance of the resulting event-specific jump. Set weight to the reciprocal of that variance, winsorized at the frozen 1st and 99th percentiles; a zero or non-estimable variance excludes that event from calibration and is reported.

[PI-DECISION 17] Calibration reporting. Search a 0.01 delta grid on $[0, 1]$, report the minimizer and percentile 95% bootstrap interval, and label estimates boundary-limited when the minimizer is 0 or 1. Do not report delta_hat if the minimum-estimability condition fails.

11. Frozen output and deviation policy

W5 must emit the full mapping $\times$ cost $\times$ delta $\times$ failure-grid panel, crossing chronology, event table, flip-rate report, EIV diagnostics, capability-only counterfactual, distributional variant, and live-vintage decomposition. No configuration may be dropped because it performs poorly.

Any post-tag change to an event, window, dose, threshold, mapping hierarchy, sample, outcome family, or estimability rule requires a dated deviation memo approved by the PI before execution. First-run results are retained and reported. The `analysis/outcomes/guard` remains fail-closed.

References and source register

- Callaway, Goodman-Bacon, and Sant'Anna (2024), continuous-treatment DiD.
- Callaway and Sant'Anna (2021), multiple-period DiD.
- Rambachan and Roth (2023), parallel-trends sensitivity.
- DAX proposal and execution plan in `docs/`.
- Signed feasibility evidence in `dax/memo/feasibility_note.md` and

`dax/memo/w05_legwork_2026-07-10.md`.

- OpenAI API [changelog](#) and

[deprecations](#), retrieved 2026-08-06.